1. Problem

One trained credit model can create different governance duties in different jurisdictions.

INPUT

Trained logistic score

predict_proba becomes model_score

→

US LENS

0.80 warning level

Adverse-action reasons + monitoring

→

EU LENS

0.90 internal escalation

Article 10/14/17 evidence package

Why this cannot be a memo or static spreadsheet

- rules need effective dates and versions
- same metric needs jurisdiction-specific interpretation
- explanation, fairness and oversight evidence must stay linked to model outputs

2. Why Citi, Why Meridian

A realistic cross-border bank case matched to a deliberately narrow RegTech product.

Why Citi

- US-headquartered global bank
- public European operating footprint
- credit governance is plausible but not alleged

Why Meridian

- 25-40 person specialist RegTech
- explainability + model-risk governance niche
- enterprise SaaS plus implementation support

Boundary: this is a plausible design case, not proof of Citi's actual model inventory or compliance posture.

3. Why The Product Fits Meridian

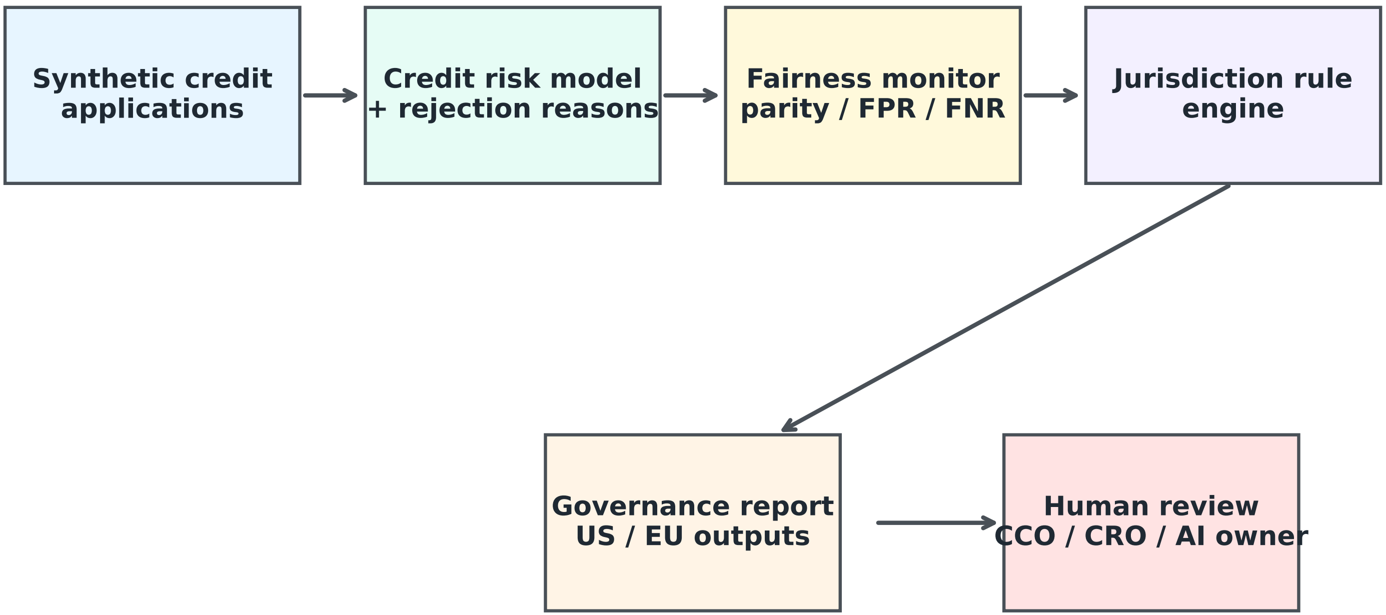
The product is valuable because it converts regulatory divergence into repeatable evidence, not because it promises legal certainty.

Explainability	linear contributions -> ranked adverse-action reasons
Fairness monitoring	approval parity + FPR/FNR + sensitivity tests
Rule engineering	versioned thresholds + conflict notes + owners
Human judgement	review bands + sign-off logs + no auto-remediation

Commercial logic: sell a narrow, auditable product to global banks that must monitor model and rule changes continuously.

4. Architecture View

Where automation stops and accountable judgement begins



Design principle: the tool calculates, records and escalates; it does not make the final credit-policy or legal-compliance judgement.

5. Rule Engine

Dimension	US Mode	EU Mode
Regulatory focus	Adverse-action reasons; fair-lending risk governance	High-risk AI; bias testing and oversight
Metric	Disparate-impact style approval parity evidence	Approval parity ratio as high-risk AI bias test
Threshold	0.80 internal warning	0.90 internal escalation not statutory
Observed parity	0.885	0.885
Tool conclusion	Monitor; do not state legal violation	Escalate EU high-risk AI governance review
Human role	Review adverse-action reason quality	Document bias testing, Article 14 oversight sign-off

Thresholds are governance settings. EU legal duties are mapped separately through Article 10, 14 and 17 controls.

6. US/EU Divergence

One model. One portfolio. Two governance conclusions.

Observed approval parity = 0.885

US mode

Threshold 0.80

Monitor;
do not state legal violation

Specific adverse-action reasons
Fair-lending risk evidence
Legal posture versioned

EU mode

Internal threshold 0.90

Escalate high-risk
AI governance review

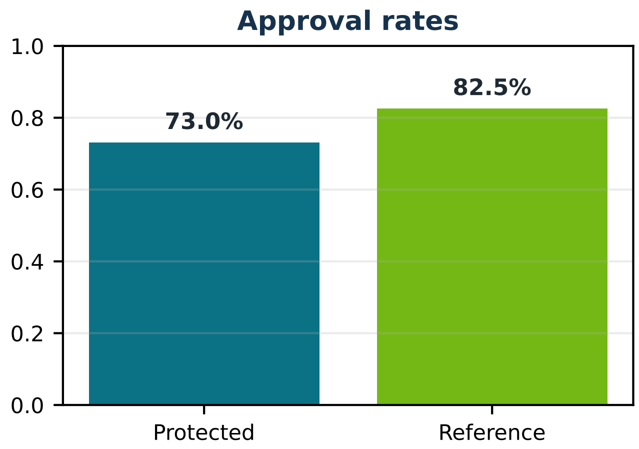
Article 10 data governance
Article 14 oversight
Article 17 quality system

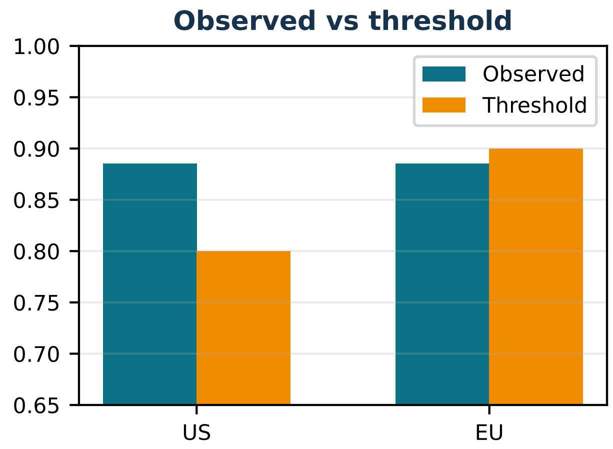
The EU threshold is not statutory; it triggers Meridian's evidence workflow for separately mapped AI Act controls.

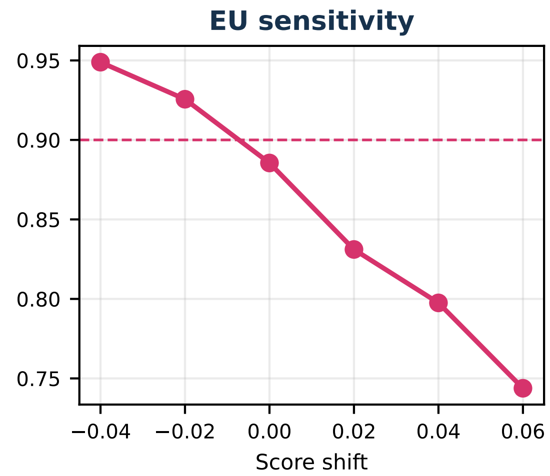
7. Dashboard Snapshot

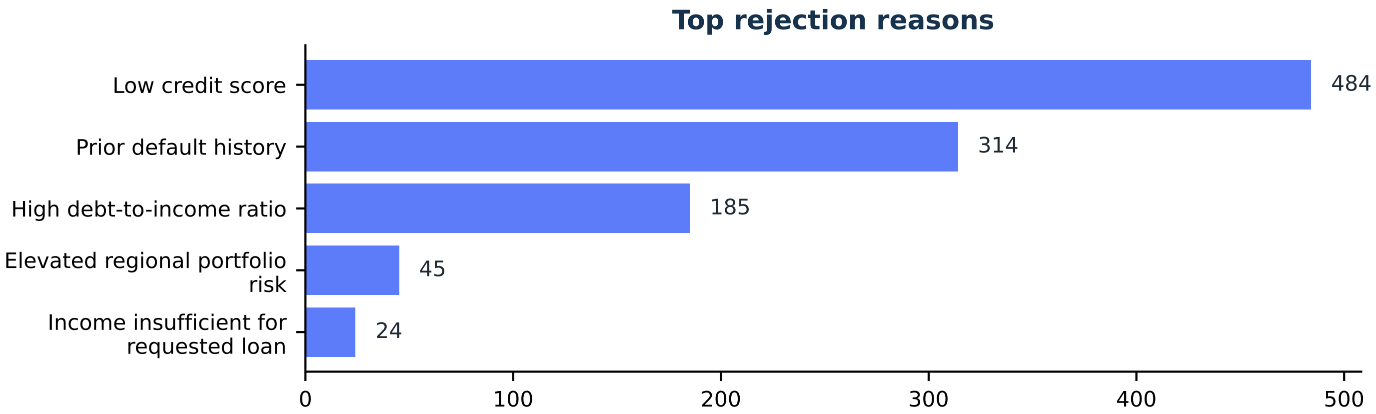
A management view should make the governance consequence legible in seconds.

APPROVAL PARITY	US MODE	EU MODE
0.885	**Monitor**	**Escalate**
Below EU internal threshold	No legal conclusion	Bias + oversight evidence









Management readout

US mode keeps monitoring; EU mode escalates because parity falls below Meridian's internal high-risk AI governance threshold.

8. Explainability Output

The tool creates consumer-facing reason evidence and separate jurisdiction-specific governance text.

applicant_id	model_score	decision_reason_1	decision_reason_2
APP-00011	0.283	High debt-to-income ratio	Elevated regional portfolio risk
APP-00013	0.254	Low credit score	Elevated regional portfolio risk
APP-00014	0.523	Low credit score	High debt-to-income ratio
APP-00024	0.440	Low credit score	High debt-to-income ratio
APP-00025	0.331	High debt-to-income ratio	Low credit score

US output example: adverse-action notice required with ranked principal reasons.

EU output example: high-risk AI documentation / human oversight evidence is required separately from customer-facing reasons.

9. Failure Modes

The dashboard must surface risk, not hide it

Key risks are rule changes mid-period, model drift, data drift, jurisdiction misconfiguration, missing protected attributes, contradictory rules and over-reliance on technical explanations. The tool responds by versioning rules, preserving effective dates, separating legal conclusions from statistical evidence and escalating unresolved choices to human owners.

10. Boundaries

What the tool does not do, and why those limits are part of good governance.

Not legal advice

Reason: statistical disparity is evidence for review, not a legal conclusion.

Not autonomous approval

Reason: credit policy and overrides require accountable human judgement.

Not auto-remediation

Reason: changing thresholds or retraining models can create new consumer harms.

Not proof of Citi compliance

Reason: the prototype uses synthetic data and public case-study evidence only.

Management message: the tool escalates evidence to humans; it does not hide policy choices inside automation.

11. Ask / Next Steps

Current demo parity 0.885; US threshold 0.80; EU internal threshold 0.90

The ask is approval to treat Meridian FairAI Governance as a narrow RegTech prototype for Citi-style cross-border AI credit governance. The next build would validate with real customer data, legal sign-off, model validation, audit logging, access control, production monitoring and integration with model inventory and issue-management workflows.